

Lattice Gas

inspired by Bryan Clark's [course](#)

Introduction

A **cellular automaton** is a computational model used to simulate complex systems that exhibit emergent behavior. The cellular automaton can be imagined as a small universe with its own internal rules, and they can be as simple or complex as we want them to be. This is one of the best examples of *emergence*, as one can observe a global pattern arising from local interactions.

We will work with a simplified model of gas particles diffusing on a lattice. Our main goal is to observe how the system evolves and to study one of the central concepts in statistical physics: **entropy**. We will quantify this by measuring the **compressibility** of the image of the lattice: configurations that are more disordered are harder to compress, and thus have higher entropy.

By running the simulation for a certain number of steps and taking snapshots at regular intervals, we can visualize how the initially ordered system becomes increasingly disordered and approaches equilibrium.

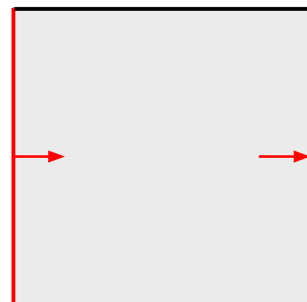
Simulation

In this lattice gas automaton, we use two colors: black and white. Black represents gas particles, and white represents empty space.

The system is initialized with the **right half** of the 100×100 lattice filled with gas (black), and the **left half** empty (white).

The update rules are:

- Randomly select a site
- Randomly select one of its neighbors
- Swap the contents of the two sites (even if they are the same)



The simulation uses **open boundary conditions**, the top and bottom edges do not wrap, whereas the left and right edges do. Care must be taken to avoid accessing out-of-bounds memory. If the boundary conditions are too difficult, focus on the general case for now -- that is, sites away from the boundary.

Tasks

Animation

1. We will call 100x100 updates (steps) one sweep. Save a snapshot of the simulation every sweep for 1500 sweeps.

Q1. *Why do we save snapshots of the configuration at time intervals of 100x100 steps?*

2. Produce an animation of the system using every 5th snapshot.

Compression File Size

1. For each snapshot, use the package `gzip` to compress the data with the following line of code `gzip.compress(snapshot.tobytes())`
2. Use the `getsizeof` function from the `sys` package to obtain the file size of each compressed snapshot.
3. Plot the size of the compressed image as a function of the number of sweeps. As always, make sure your plot has appropriate labels and titles.

Q2. *Why are more “disordered” images harder to compress?*

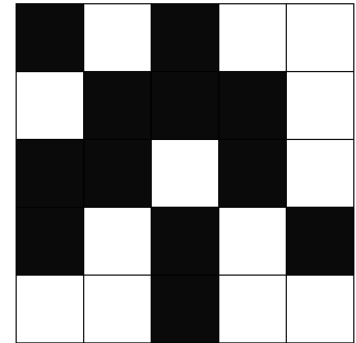
Q3. *What file size value does your plot converge to? Do you expect this value to change if you change the initial configuration of particles? Why or why not?*

[Optional] Coarse-Graining and Direct Entropy Calculation

1. Create a 1x100 array where each entry is the average number of particles in that column of gas. For example, for a 5x5 grid this is given by the figure.

Q4. How many microstates of the 100x100 fine-grained lattice correspond to the initial configuration in this 25x25 coarse-grained lattice?

2. This array represents the density of the gas at each location x . Create an animation that shows how this density profile changes over time.
3. Calculate the entropy of the density profile at each time using `scipy.stats.entropy` and create a plot of the entropy v.s. time.
 - a. *Alternatively*, you can calculate this by hand by first normalizing your density profile and using the formula for the Shannon entropy:



0.6	0.4	0.8	0.4	0.2
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$$H[\{p_i\}] = - \sum_i p_i \log p_i$$